Supplementary

Table S1. Predictive performance and computational efficiency of the evaluated models when treating chol = 0 and trestbps = 0 as missing values.

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
AdaBoost	0.7392 ± 0.0632	0.7656 ± 0.0393	0.8301 ± 0.1277	0.7212 ± 0.0374	0.7892 ± 0.0858	0.2217 ± 0.0100	0.4920 ± 0.2116	0.0355 ± 0.0199	0.1816 ± 0.1384
CNN	0.7138 ± 0.0744	0.7565 ± 0.1144	0.7201 ± 0.1920	0.8171 ± 0.0253	0.6771 ± 0.1063	0.2093 ± 0.0233	38.1409 ± 19.2544	0.0020 ± 0.0012	0.0086 ± 0.0037
CatBoost	0.7499 ± 0.0419	0.7825 ± 0.0393	0.8072 ± 0.1352	0.7743 ± 0.0505	0.7971 ± 0.0793	0.1737 ± 0.0237	0.9279 ± 0.8087	0.0007 ± 0.0002	0.0036 ± 0.0020
CatBoost-Native	0.7424 ± 0.0218	0.7765 ± 0.0523	0.7876 ± 0.1465	0.7816 ± 0.0391	0.7924 ± 0.0801	0.1765 ± 0.0226	11.8975 ± 1.4048	0.0017 ± 0.0004	0.0080 ± 0.0015
GCN-Inductive	0.7573 ± 0.0216	0.7859 ± 0.0762	0.7894 ± 0.1626	0.8003 ± 0.0467	0.7954 ± 0.0641	0.1761 ± 0.0159	0.7813 ± 0.4538	0.0087 ± 0.0035	0.0382 ± 0.0067
GCN-Transductive	0.7124 ± 0.2234	0.7478 ± 0.1978	0.7151 ± 0.2349	0.7946 ± 0.1442	0.6950 ± 0.1857	0.1921 ± 0.1143	1.2773 ± 1.3410	0.0087 ± 0.0029	0.0397 ± 0.0125
Logistic Regression	0.7502 ± 0.0320	0.7911 ± 0.0507	0.7798 ± 0.1541	0.8257 ± 0.0654	0.8037 ± 0.0812	0.1783 ± 0.0218	0.0166 ± 0.0158	0.0001 ± 0.0001	0.0005 ± 0.0002

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
Random Forest	0.7274 ± 0.0333	0.7727 ± 0.0452	0.7686 ± 0.1606	0.8046 ± 0.0768	0.7880 ± 0.0771	0.1834 ± 0.0140	0.4689 ± 0.1756	0.0304 ± 0.0160	0.1516 ± 0.1022
SVM	0.7501 ± 0.0103	0.7886 ± 0.0672	0.7724 ± 0.1704	0.8306 ± 0.0527	0.7983 ± 0.0587	0.1772 ± 0.0089	0.0719 ± 0.0433	0.0063 ± 0.0035	0.0294 ± 0.0143
XGBoost	0.6971 ± 0.0458	0.7441 ± 0.0325	0.7619 ± 0.1621	0.7571 ± 0.0917	0.7566 ± 0.0709	0.2412 ± 0.0270	0.3361 ± 0.1404	0.0009 ± 0.0001	0.0043 ± 0.0022

Values are reported as mean ± standard deviation. Lower Brier scores and shorter computational times indicate better performance. Bold values indicate the best mean value for each metric.

Table S2. Predictive performance and computational efficiency of the evaluated models after adding binary missing-value indicators.

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
AdaBoost	0.7310 ± 0.0630	0.7591 ± 0.0508	0.8183 ± 0.1492	0.7266 ± 0.0723	0.7838 ± 0.1019	0.2242 ± 0.0098	0.3328 ± 0.0130	0.0193 ± 0.0009	0.0945 ± 0.0380
CNN	0.6927 ± 0.2049	0.7420 ± 0.1905	0.6732 ± 0.2209	0.8544 ± 0.1719	0.5511 ± 0.1932	0.2373 ± 0.1394	5.5706 ± 0.7168	0.0010 ± 0.0001	0.0047 ± 0.0018

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
CatBoost	0.7660 ± 0.0636	0.7971 ± 0.0399	0.8280 ± 0.1206	0.7799 ± 0.0525	0.7880 ± 0.1015	0.1713 ± 0.0315	0.4908 ± 0.1257	0.0006 ± 0.0001	0.0027 ± 0.0010
CatBoost-Native	0.7557 ± 0.0644	0.7802 ± 0.0494	0.8296 ± 0.1113	0.7432 ± 0.0454	0.7761 ± 0.1017	0.1738 ± 0.0330	8.4339 ± 0.1378	0.0009 ± 0.0001	0.0041 ± 0.0012
GCN-Inductive	0.7557 ± 0.0182	0.7821 ± 0.0776	0.7914 ± 0.1400	0.7928 ± 0.1053	0.6964 ± 0.1839	0.1856 ± 0.0342	0.3956 ± 0.0286	0.0064 ± 0.0015	0.0299 ± 0.0096
GCN-Transductive	0.7473 ± 0.0380	0.7766 ± 0.0682	0.7906 ± 0.1183	0.7833 ± 0.1468	0.7621 ± 0.1203	0.1982 ± 0.0348	0.8460 ± 0.8933	0.0071 ± 0.0019	0.0323 ± 0.0064
Logistic Regression	0.7471 ± 0.0379	0.7773 ± 0.0541	0.8106 ± 0.1404	0.7757 ± 0.1317	0.8018 ± 0.0896	0.1800 ± 0.0214	0.0124 ± 0.0123	0.0001 ± 0.0000	0.0004 ± 0.0001
Random Forest	0.7426 ± 0.0360	0.7801 ± 0.0393	0.7898 ± 0.1370	0.7892 ± 0.0697	0.7779 ± 0.1038	0.1774 ± 0.0174	0.3184 ± 0.0156	0.0178 ± 0.0004	0.0877 ± 0.0381
SVM	0.6956 ± 0.1541	0.7731 ± 0.1272	0.6907 ± 0.2187	0.9287 ± 0.0555	0.7569 ± 0.0956	0.1925 ± 0.0396	0.0380 ± 0.0091	0.0039 ± 0.0010	0.0174 ± 0.0025
XGBoost	0.7212 ± 0.0526	0.7547 ± 0.0559	0.7820 ± 0.1442	0.7448 ± 0.0612	0.7565 ± 0.0918	0.2193 ± 0.0443	0.1427 ± 0.0911	0.0007 ± 0.0001	0.0035 ± 0.0013

Table S3. Predictive performance and computational efficiency of the evaluated models after removing ca, thal, and slope.

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
AdaBoost	0.7513 ± 0.0467	0.7797 ± 0.0426	0.8173 ± 0.1230	0.7558 ± 0.0439	0.7871 ± 0.0794	0.2201 ± 0.0055	0.4529 ± 0.1583	0.0387 ± 0.0203	0.2038 ± 0.1694
CNN	0.7786 ± 0.0739	0.7939 ± 0.1111	0.7775 ± 0.1351	0.8132 ± 0.0890	0.7521 ± 0.0798	0.1628 ± 0.0482	38.0934 ± 7.6911	0.0021 ± 0.0015	0.0086 ± 0.0034
CatBoost	0.7440 ± 0.0464	0.7792 ± 0.0364	0.7977 ± 0.1225	0.7735 ± 0.0462	0.7868 ± 0.0866	0.1762 ± 0.0264	0.6165 ± 0.1298	0.0008 ± 0.0002	0.0039 ± 0.0020
CatBoost-Native	0.7580 ± 0.0311	0.7880 ± 0.0495	0.8045 ± 0.1258	0.7826 ± 0.0409	0.8012 ± 0.0718	0.1694 ± 0.0202	11.9027 ± 1.5104	0.0013 ± 0.0002	0.0062 ± 0.0015
GCN-Inductive	0.7846 ± 0.0748	0.8032 ± 0.0851	0.8075 ± 0.1097	0.8051 ± 0.0916	0.7690 ± 0.1056	0.1586 ± 0.0444	0.8222 ± 0.4054	0.0103 ± 0.0030	0.0458 ± 0.0043
GCN-Transductive	0.7485 ± 0.0394	0.7802 ± 0.0543	0.7929 ± 0.1246	0.7760 ± 0.0161	0.7567 ± 0.1112	0.1730 ± 0.0301	1.2107 ± 1.1127	0.0090 ± 0.0036	0.0393 ± 0.0047
Logistic Regression	0.7500 ± 0.0582	0.7833 ± 0.0365	0.8095 ± 0.1094	0.7704 ± 0.0692	0.7983 ± 0.0748	0.1737 ± 0.0357	0.0122 ± 0.0069	0.0001 ± 0.0001	0.0006 ± 0.0005
Random Forest	0.7702 ±	0.8034 ±	0.8082 ±	0.8096 ±	0.7956 ±	0.1665 ±	0.4316 ±	0.0309 ± 0.0157	0.1615 ± 0.1273

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
	0.0411	0.0418	0.1144	0.0565	0.0829	0.0202	0.1502		
SVM	0.7806 ± 0.0595	0.8065 ± 0.0813	0.7850 ± 0.1202	0.8347 ± 0.0571	0.8015 ± 0.0715	0.1524 ± 0.0349	0.0527 ± 0.0282	0.0057 ± 0.0029	0.0267 ± 0.0140
XGBoost	0.7203 ± 0.0386	0.7575 ± 0.0534	0.7760 ± 0.1544	0.7574 ± 0.0515	0.7784 ± 0.0709	0.2125 ± 0.0241	0.2985 ± 0.1696	0.0009 ± 0.0001	0.0044 ± 0.0022

Table S4. Predictive performance and computational efficiency of the evaluated models after hyperparameter tuning using Grid Search.

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
AdaBoost	0.7295 ± 0.0563	0.7673 ± 0.0279	0.7978 ± 0.1361	0.7610 ± 0.0934	0.7926 ± 0.0816	0.1957 ± 0.0163	0.2193 ± 0.1326	0.0127 ± 0.0070	0.0723 ± 0.0619
CNN	0.6460 ± 0.1243	0.6874 ± 0.1242	0.7504 ± 0.2136	0.6991 ± 0.2059	0.6735 ± 0.0922	0.2462 ± 0.0775	5.0904 ± 0.4399	0.0010 ± 0.0001	0.0046 ± 0.0017
CatBoost	0.7538 ± 0.0704	0.7814 ± 0.0371	0.8334 ± 0.1023	0.7418 ± 0.0401	0.7881 ± 0.0981	0.1734 ± 0.0367	0.3752 ± 0.0688	0.0006 ± 0.0000	0.0027 ± 0.0010
CatBoost-Native	0.7627 ± 0.0390	0.7861 ± 0.0529	0.8226 ± 0.1193	0.7586 ± 0.0039	0.7769 ± 0.0988	0.1743 ± 0.0305	10.2732 ± 2.2019	0.0011 ± 0.0006	0.0049 ± 0.0015

Model	Accuracy	F1-score	Precision	Recall	AUC	Brier score	Training time (s)	Inference time/batch (s)	Inference time/patient (ms)
GCN-Inductive	0.7806 ± 0.0694	0.7881 ± 0.0901	0.8377 ± 0.0983	0.7503 ± 0.1099	0.7747 ± 0.1132	0.1617 ± 0.0373	0.3926 ± 0.0196	0.0062 ± 0.0014	0.0287 ± 0.0077
GCN-Transductive	0.8037 ± 0.0750	0.8082 ± 0.0976	0.8410 ± 0.0805	0.7862 ± 0.1369	0.8299 ± 0.0785	0.1378 ± 0.0376	0.3933 ± 0.0220	0.0073 ± 0.0020	0.0330 ± 0.0060
Logistic Regression	0.7311 ± 0.0309	0.7677 ± 0.0536	0.7836 ± 0.1582	0.7729 ± 0.0549	0.7943 ± 0.0809	0.1826 ± 0.0231	0.0043 ± 0.0015	0.0001 ± 0.0000	0.0004 ± 0.0001
Random Forest	0.7402 ± 0.0491	0.7804 ± 0.0362	0.7878 ± 0.1356	0.7921 ± 0.0733	0.8017 ± 0.0840	0.1736 ± 0.0158	0.2778 ± 0.0194	0.0160 ± 0.0008	0.0786 ± 0.0322
SVM	0.7519 ± 0.0337	0.7858 ± 0.0580	0.7882 ± 0.1513	0.8006 ± 0.0469	0.7996 ± 0.0863	0.1635 ± 0.0250	0.0365 ± 0.0096	0.0041 ± 0.0011	0.0184 ± 0.0027
XGBoost	0.7479 ± 0.0566	0.7753 ± 0.0427	0.8187 ± 0.1146	0.7444 ± 0.0388	0.7988 ± 0.0950	0.1697 ± 0.0358	0.0493 ± 0.0037	0.0007 ± 0.0000	0.0033 ± 0.0014

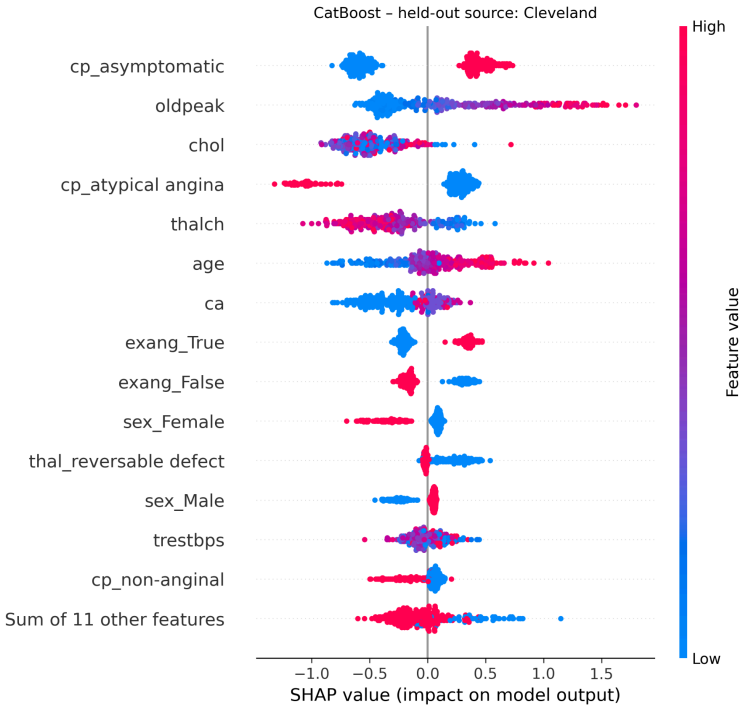


Figure S1 SHAP beeswarm plot for CatBoost with Cleveland as the held-out source.

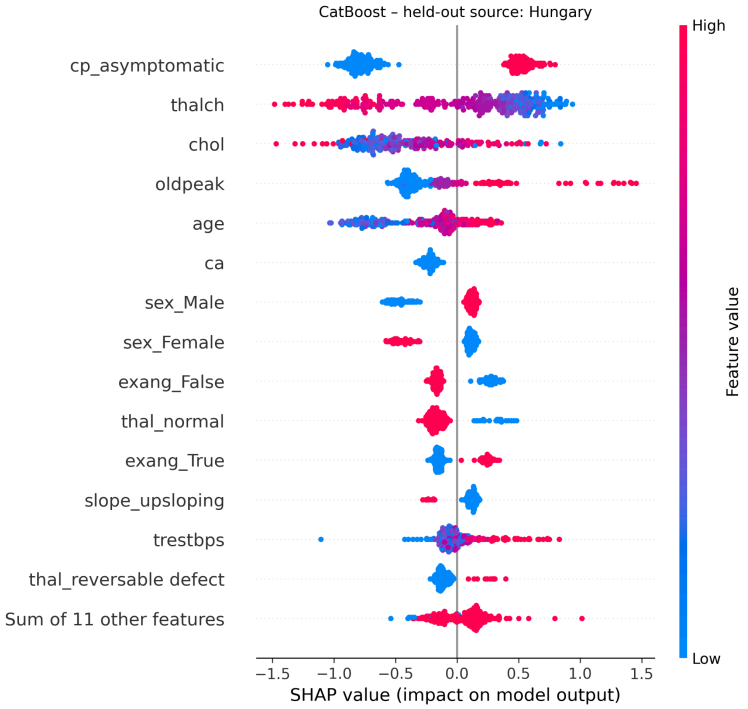


Figure S2 SHAP beeswarm plot for CatBoost with Hungary as the held-out source.

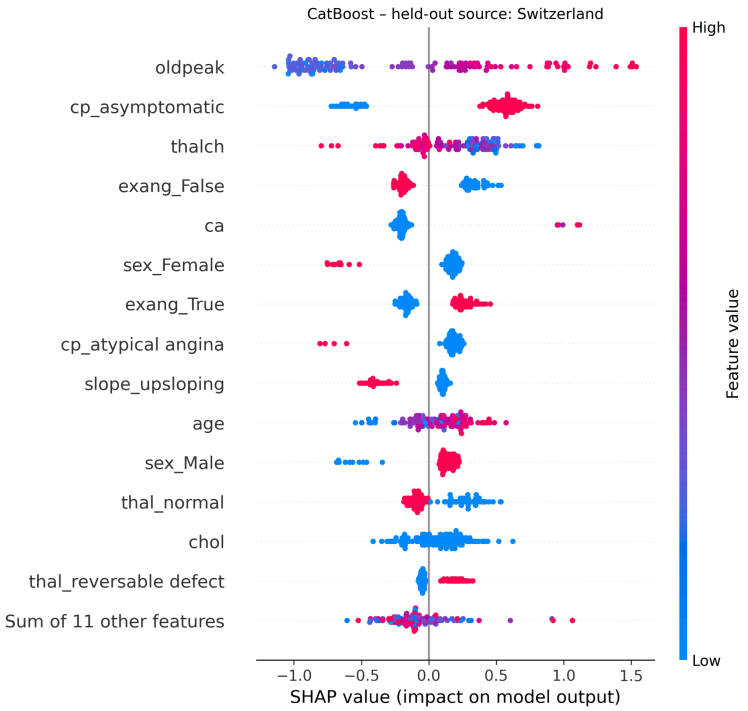


Figure S3 SHAP beeswarm plot for CatBoost with Switzerland as the held-out source.

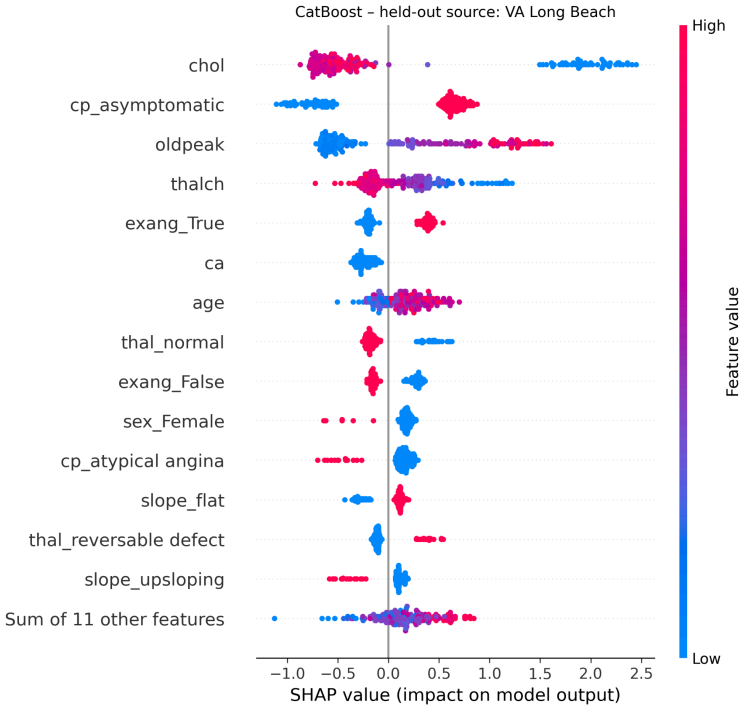


Figure S4 SHAP beeswarm plot for CatBoost with VA Long Beach as the held-out source.

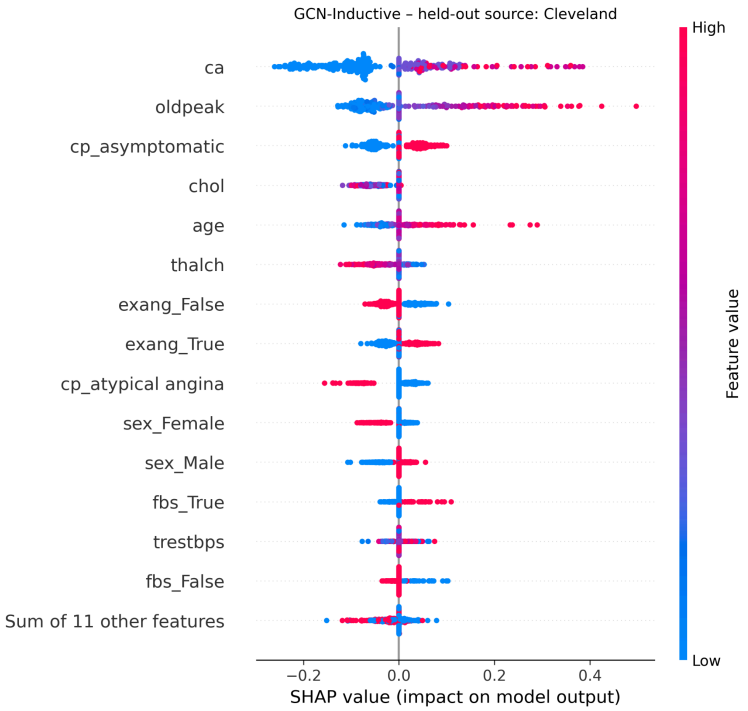


Figure S5 SHAP beeswarm plot for GCN-Inductive with Cleveland as the held-out source.

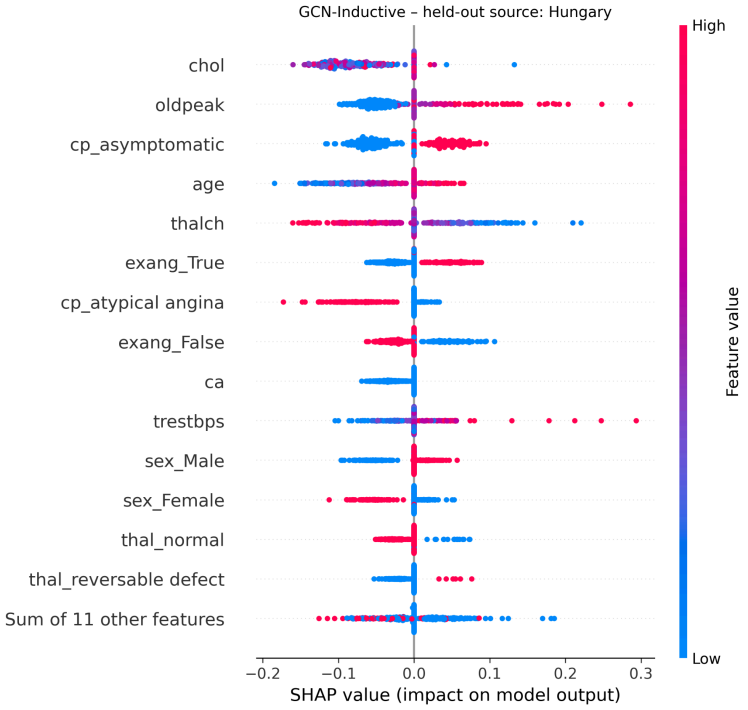


Figure S6 SHAP beeswarm plot for GCN-Inductive with Hungary as the held-out source.

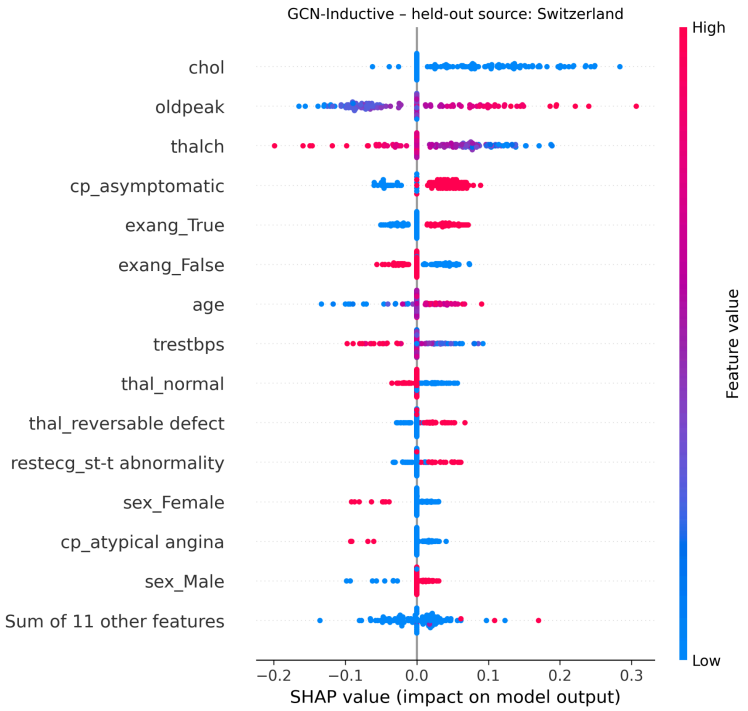


Figure S7 SHAP beeswarm plot for GCN-Inductive with Switzerland as the held-out source.

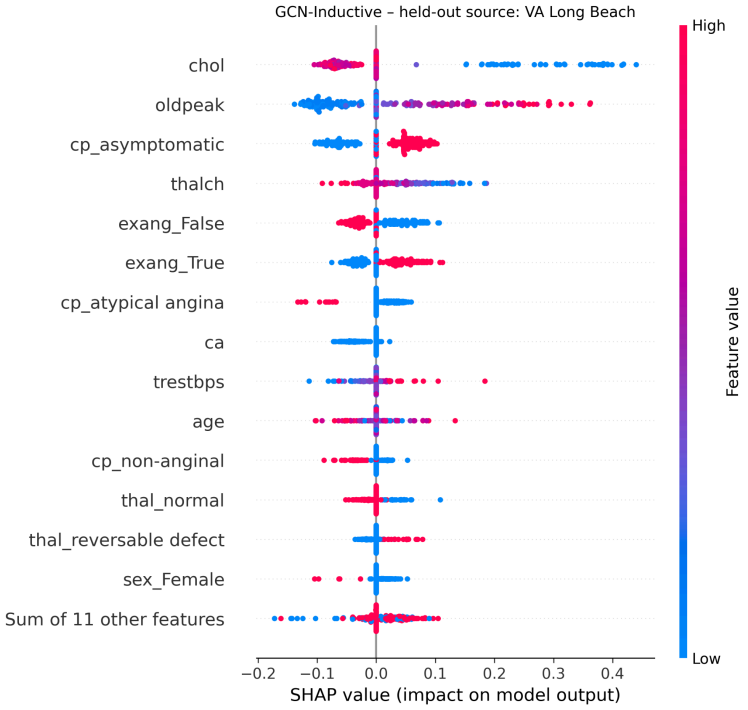


Figure S8 SHAP beeswarm plot for GCN-Inductive with VA Long Beach as the held-out source.